

Sample Question:

A digital signature provides for data integrity and a MAC provides for data integrity. Why does a digital signature also provides for non-repudiation while a MAC does not?

Answer:

- **MAC (Message Authentication Code):** এটি **সিমেট্রিক কি (Symmetric Key)** ব্যবহার করে। এর মানে হলো, প্রেরক (Alice) এবং প্রাপক (Bob) দুজনের কাছেই একই গোপন চাবি (Shared Secret Key) থাকে। যেহেতু দুজনের কাছেই একই চাবি আছে, তাই Bob চাইলেই Alice-এর নামে একটি ভুয়া মেসেজ তৈরি করতে পারে। যদি Alice পরে অস্বীকার করে যে সে মেসেজটি পাঠায়নি, তবে Bob তৃতীয় কারো কাছে প্রমাণ করতে পারবে না যে এটি Alice-ই পাঠিয়েছে, কারণ Bob নিজেও সেটি তৈরি করার ক্ষমতা রাখে। তাই এতে **নন-রেপুডিয়েশন** নেই।
- **ডিজিটাল সিগনেচার (Digital Signature):** এটি **পাবলিক কি (Public Key)** ক্রিপ্টোগ্রাফি ব্যবহার করে। এখানে Alice তার নিজস্ব **প্রাইভেট কি (Private Key)** দিয়ে মেসেজ সাইন করে, যা পৃথিবীতে আর কারো কাছে নেই। Bob বা অন্য যে কেউ Alice-এর **পাবলিক কি** দিয়ে তা যাচাই করতে পারে, কিন্তু কেউ সেটা জাল করতে পারে না। যেহেতু এই সিগনেচার শুধুমাত্র Alice-এর চাবি দিয়েই সম্ভব, তাই সে পরে অস্বীকার করতে পারে না যে মেসেজটি সে পাঠায়নি। একেই বলে **নন-রেপুডিয়েশন**।

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- **MAC (Message Authentication Code):**
It uses a **symmetric key**, which means both the sender (Alice) and the receiver (Bob) share the same secret key. Since both parties have the same key, Bob could create a fake message pretending to be Alice. If Alice later denies sending the message, Bob cannot prove to a third party that it was actually sent by Alice, because Bob himself has the ability to generate it. Therefore, **MAC does not provide non-repudiation.**
 - **Digital Signature:**
It uses **public key cryptography**. Here, Alice signs the message using her own **private key**, which no one else possesses. Bob or anyone else can verify the signature using Alice's **public key**, but no one can forge it. Since this signature can only be created using Alice's private key, she cannot later deny sending the message. This property is called **non-repudiation.**

Sample Question:

Consider the elliptic curve $E : y^2 = x^3 + 7x + b \pmod{11}$.

- Determine b so that the point $P = (4,5)$ is on the curve E .
- Using the b found in part a, list all points on E .
- Using the b found in part a, find the sum $(4,5) + (5,4)$ on E .
- Using the b found in part a, find the point $3(4,5)$.

Answer:

Given:

Elliptic Curve:

$$\curvearrowright E : y^2 = x^3 + 7x + b \pmod{11}$$

$$E: y^2 = x^3 + 7x + b \pmod{11}$$

Point:

$$\curvearrowright P = (4, 5)$$

✓ (a) Find b

Point (4,5) curve এ বসাই:

$\curvearrowright$ Left side:

$$y^2 = 5^2 = 25 \equiv 3 \pmod{11}$$

$\curvearrowright$ Right side:

$$x^3 + 7x + b$$

$$= 4^3 + 7 \times 4 + b$$

$$= 64 + 28 + b$$

$$= 92 + b$$

$\curvearrowright$ mod 11:

$$92 \equiv 4$$

So, RHS = 4 + b

$\curvearrowright$ Equation:

$$3 = 4 + b \pmod{11}$$

$$\curvearrowright b = 3 - 4 = -1 \equiv 10 \pmod{11}$$

✓ Answer: b=10

(b) Using the b found in part a, list all points on E

Find all points on:

$$\curvearrowright y^2 = x^3 + 7x + 10 \pmod{11}$$

$\curvearrowright$ Step 1: আগে mod 11 এ square values (y^2) কী কী হতে পারে সেটা লিখে রাখো

y	$y^2 \pmod{11}$
0	0
1	1

y	$y^2 \bmod 11$
2	4
3	9
4	5
5	3
6	3
7	5
8	9
9	4
10	1

Step 2: এখন x বসাই ($0 \rightarrow 10$)

আমরা শুধু check করবো RHS এই set এর মধ্যে পড়ে কিনা
 $y^2 = x^3 + 7x + 10 \pmod{11}$

✓ $x = 0$

RHS = 10 ✗ (not in list)

✓ $x = 1$

RHS = 7 ✗

✓ $x = 2$

RHS = 10 ✗

✓ $x = 3$

RHS = 3 ✓

☞ $y^2 = 3 \rightarrow y = 5, 6$

✓ (3,5), (3,6)

✓ $x = 4$

RHS = 3 ✓

☞ $y = 5, 6$

✓ (4,5), (4,6)

✓ $x = 5$

RHS = 5 ✓

☞ $y = 4, 7$

✓ (5,4), (5,7)

✓ $x = 6$

RHS = 4 ✓

☞ **y = 2, 9**

✓ (6,2), (6,9)

✓ **x = 7**

RHS = 6 ✗

✓ **x = 8**

RHS = 6 ✗

✓ **x = 9**

RHS = 10 ✗

✓ **x = 10**

RHS = 2 ✗

✓ Final Answer (Very Important)

☞ All points:

✓ (3,5), (3,6)

✓ (4,5), (4,6)

✓ (5,4), (5,7)

✓ (6,2), (6,9)

☞ + Infinity point (O)

✓ Problem (c)

Using the b found in part a, find the sum $(4,5) + (5,4)$ on E.

🔥 Step 1: Formula (Different point addition)

👉 $m = \frac{y_2 - y_1}{x_2 - x_1}$

👉 $x_3 = m^2 - x_1 - x_2$

👉 $y_3 = m(x_1 - x_3) - y_1$

🔥 Step 2: m (slope) বের করি

👉 $m = (4 - 5)/(5 - 4) = -1/1 = -1$

👉 mod 11 এ:

👉 $-1 \equiv 10$

✅ so, $m = 10$

🔥 Step 3: x_3 বের করি

👉 $x_3 = 10^2 - 4 - 5$

👉 $= 100 - 9 = 91$

👉 mod 11:

👉 $91 \equiv 3$

✅ $x_3 = 3$



🔥 Step 4: y_3 বের করি

👉 $y_3 = 10(4 - 3) - 5$

👉 $= 10 \times 1 - 5 = 5$

✅ $y_3 = 5$

✅ Final Answer

👉 $(4,5) + (5,4) = (3,5)$

d. Using the 6 found in part a, find the point 3(4,5).

✓ Problem (d)

Find:

👉 $3(4, 5)$

🔥 Main Idea

👉 $3P = 2P + P$

👉 So first বের করতে হবে:

1. $2P$
2. তারপর $(2P + P)$

✓ Step 1: Find $2P = (4,5) + (4,5)$

👉 Same point → Doubling formula

Formula:

👉 $m = \frac{3x^2 + a}{2y}$

👉 এখানে $a = 7$

m বের করি:

👉 $m = (3 \times 4^2 + 7) / (2 \times 5)$

👉 $= (48 + 7) / 10 = 55 / 10$



Ekhane m er man ber korar jonno ei sutrota lagbe

$$a \cdot x \equiv 1 \pmod{p}$$

$$55 \bmod 11 = 0$$

but 10 er jonno sutrota apply hobe

tar mane 10 er sathe koto gun korle and tar sathe 11 mod korle = 1 hobe? Eta mainly concept

$$10 \cdot 10 \bmod 11 = 1$$

$$\Rightarrow 10^{-1} = 10$$

$$\text{Tar mane } m = 0 \cdot 10^{-1} = 0 \cdot 10 = 0$$

Evabei m er man ta ber koreche

aro ekta example dilam

$$56/14 \rightarrow 1/3 \rightarrow 1 \times 3^{-1} \rightarrow 1 \times 4 = 4$$

mod 11 এ simplify:

👉 $55 \equiv 0$

👉 10 এর inverse = 10

👉 so:

✓ $m = 0 \times 10 = 0$

🔥 এখন x_3 :

👉 $x_3 = m^2 - 2x$

👉 $= 0 - 8 = -8 \equiv 3$

🔥 এখন y_3 :

👉 $y_3 = m(x - x_3) - y$

👉 $= 0 - 5 = -5 \equiv 6$

✓ $2P = (3,6)$

✓ **Step 2: Find $3P = (3,6) + (4,5)$**

👉 Different point $\rightarrow$ normal formula

m বের করি:

👉 $m = (5 - 6)/(4 - 3) = -1 \equiv 10$

🔥 x_3 :

👉 $x_3 = 10^2 - 3 - 4$

👉 $= 100 - 7 = 93 \equiv 5$

🔥 y_3 :

👉 $y_3 = 10(3 - 5) - 6$

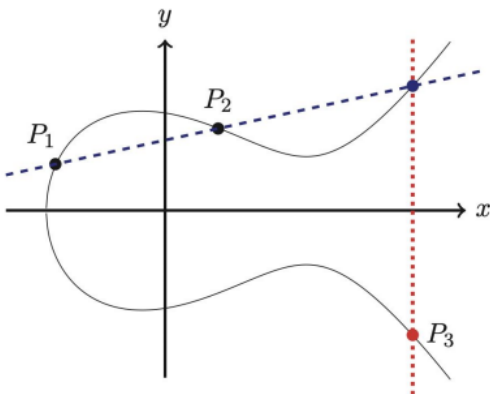
👉 $= -20 - 6 = -26 \equiv 7$

✅ **Final Answer**

👉 $3(4,5) = (5,7)$

Question: What is an Elliptic Curve?

Answer: An elliptic curve E is the graph of an equation of the form $y^2 = x^3 + ax + b$ □ Also includes a “point at infinity”



□ Consider elliptic curve

$$E: y^2 = x^3 - x + 1$$

□ If P_1 and P_2 are on E , we can define addition,

$$P_3 = P_1 + P_2$$

as shown in picture

□ Addition is all we need...

What is Man-in-the-Middle (MITM) Attack

A Man-in-the-Middle (MITM) attack is a type of security attack in which an attacker secretly intercepts, relays, and possibly alters the communication between two parties who believe they are communicating directly with each other

In simple terms:

- o Alice thinks she is talking to Bob.
- o Bob thinks he is talking to Alice.
- o But Trudy (the attacker) is silently sitting in between.

Why MITM Attacks Are Dangerous

- ❑ MITM attacks can violate all three pillars of security:

Security Property	How MITM Breaks It
Confidentiality	Attacker reads sensitive data
Integrity	Attacker alters messages
Authentication	Attacker impersonates users

Common Types of MITM Attacks

- ❑ Passive MITM
 - o Attacker only listens.
 - o No modification.
 - o Hard to detect.
- ❑ Active MITM
 - o Attacker modifies messages.
 - o Can inject fake data.
 - o More destructive.

How to Prevent MITM Attacks

❑ Cryptographic Countermeasures

Technique	Purpose
Digital certificates	Verify identity
Digital signatures	Ensure authenticity
TLS / HTTPS	Secure authenticated channels
Public Key Infrastructure (PKI)	Trust management

● What is Public Key Infrastructure (PKI)?

☞ **Public Key Infrastructure (PKI)** is a system/framework that uses **public key cryptography** to ensure secure communication.

● Symmetric Key Cryptography (Definition)

☞ **Symmetric key cryptography** is a type of encryption where the same secret key is used for both encryption and decryption of data.

● Public Key Cryptography (Definition)

☞ **Public key cryptography** is a type of encryption that uses a pair of keys—a public key for encryption and a private key for decryption.

Question: What is the primary purpose of the Diffie-Hellman (DH) algorithm? Can it be used for encryption or digital signatures?

Answer: The purpose of DH is **Key Exchange**—to establish a shared symmetric key over an insecure channel. It is **not** designed for encryption or digital signing.

Question: What mathematical problem is the security of Diffie-Hellman based on?

Answer: It is based on the **Discrete Logarithm Problem** (Given $g^k \pmod{p}$, it is computationally hard to find the exponent k).

Question: Explain the Man-in-the-Middle (MITM) attack on Diffie-Hellman. How can it be prevented?

Answer: In an MITM attack, an attacker (Trudy) intercepts the messages between Alice and Bob. Alice thinks she is talking to Bob, and Bob thinks he is talking to Alice, but both are actually sharing keys with Trudy. It can be prevented using **Digital Signatures, Public Key Certificates, or PKI**.

Question: Why is ECC considered more "efficient" than RSA?

Answer: ECC provides the same level of security as RSA but with **much smaller key sizes**. For example, a 160-bit ECC key is as secure as a 1024-bit RSA key. This results in faster processing and lower power consumption.